Chapter 9. Analysis of Variance

One-way and two-way layouts

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9.1 What is analysis of variance?

Motivation. ANOVA asks whether a characteristic differs across conditions by splitting total variation into factor and error parts.

Comparing tensile strengths by production method, a difference may come from the method or from other sources (worker skill). Repeating each method splits the variation into a between-method part and a part from other sources; if the former is large relative to the latter, the method matters.

Analysis of variance (ANOVA, R. A. Fisher), the main tool of the **design of experiments** (DOE), decomposes the sum of squares into parts attributable to the factors and compares each factor's *mean square* (sum of squares over d.f.) with the error mean square. Here

$$\text{total sum of squares} = \text{sum of squares due to the factor} + \text{sum of squares due to error.}$$

ANOVA generalizes the two-sample test (Chapter 6) to three or more means. We treat *fixed* factors: one-way ANOVA (equal and unequal replication) and two-way ANOVA (without and with replication).

9.2 One-way analysis of variance

Motivation. One factor at $k \geq 3$ levels: are the k treatment means equal?

In a **one-way layout** factor A has k levels (treatments), treatment i replicated n_i times (typically $k = 3$–5, $n_i = 3$–10), all runs in random order: a **completely randomized design** (e.g. 15 numbered units for methods A, B, C with 5 runs each, ordered by lot). Statistically, these are independent random samples from k populations; $k = 2$ is the two-sample comparison.

9.2.1 Equal replication

With n replicates per treatment the data are y_{ij} ($i = 1, \ldots, k$ treatments, $j = 1, \ldots, n$), with treatment means and grand mean

$$\bar{y}_{i\cdot} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{kn}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} \qquad (i = 1, \ldots, k).$$

Under treatment i, $Y_{ij} = \mu_i + e_{ij}$ with $e_{ij} \sim N(0, \sigma^2)$ independent. With overall mean $\mu = \frac{1}{k}\sum \mu_i$, $\alpha_i = \mu_i - \mu$ is the ith **treatment effect**, $\sum \alpha_i = 0$.

Definition (One-way ANOVA model (equal replication))

$$Y_{ij} = \mu + \alpha_i + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{k} \alpha_i = 0,$$

$i = 1, \ldots, k$, $j = 1, \ldots, n$. Equivalently:
(i) linearity $E(e_{ij}) = 0$, i.e. $E(Y_{ij}) = \mu + \alpha_i$;
(ii) equal variance $\text{Var}(e_{ij}) = \sigma^2 > 0$;
(iii) independence of the e_{ij};
(iv) normality $e_{ij} \sim N(0, \sigma^2)$.

Testing whether the treatment effects differ is the same as testing whether the k population means differ. Decompose the total deviation:

$$y_{ij} - \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{i\cdot}) + (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}). \tag{9.1}$$

$y_{ij} - \bar{y}_{i\cdot}$ is the **residual** and $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ the **deviation of the treatment effect**. Squaring and summing decomposes the **total sum of squares** SST into the **treatment** (**between**) **sum of squares** SS_{tr} and the **residual** (**within**) **sum of squares** SSE.

Proposition (Decomposition of the total sum of squares, one-way)

With $N = kn$ observations,

$$\underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\bar{y}_{\cdot\cdot})^2}_{\mathrm{SST},\ N-1} = \underbrace{n\sum_{i=1}^{k}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2}_{\mathrm{SS}_{tr},\ k-1} + \underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\bar{y}_{i\cdot})^2}_{\mathrm{SSE},\ N-k}.$$

Proof. [Supplement] Square (9.1) and sum over i, j. The cross term is

$$2\sum_{i=1}^{k}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})\sum_{j=1}^{n}(y_{ij}-\bar{y}_{i\cdot}) = 0,$$

because the inner sum is $n\bar{y}_{i\cdot} - n\bar{y}_{i\cdot} = 0$. The middle square does not depend on j, so it contributes $n\sum_i(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2$. As for degrees of freedom: SST has $N-1$ (the deviations sum to 0), SS_{tr} has $k-1$ (k treatment means minus one constraint), and SSE has $N-k$ ($n-1$ in each of k groups); indeed $N-1 = (k-1)+(N-k)$. ■

A sum of squares divided by its degrees of freedom is a **mean square**:

$$\mathrm{MS}_{tr} = \frac{\mathrm{SS}_{tr}}{k-1}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k}.$$

We test

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \alpha_i \neq 0,$$

which is the same as $H_0 : \mu_1 = \cdots = \mu_k$. If the treatments matter, SS_{tr} takes a large share of SST, so large values of $F = \mathrm{MS}_{tr}/\mathrm{MSE}$ are evidence against H_0.

Proposition (Expected mean squares, one-way)

$E(\mathrm{MSE}) = \sigma^2$ and $E(\mathrm{MS}_{tr}) = \sigma^2 + \dfrac{n}{k-1}\sum_{i=1}^{k}\alpha_i^2$. Under H_0, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. For each i, $Y_{i1}, \ldots, Y_{in}$ is a random sample from $N(\mu_i, \sigma^2)$, so its sample variance has mean σ^2: $E\sum_j(Y_{ij}-\bar{Y}_{i\cdot})^2 = (n-1)\sigma^2$. Summing over i, $E(\mathrm{SSE}) = k(n-1)\sigma^2 = (N-k)\sigma^2$.

Step 2. The treatment means $\bar{Y}_{i\cdot} \sim N(\mu+\alpha_i, \sigma^2/n)$ are independent. For independent Z_i with means m_i and common variance v, $E\sum(Z_i-\bar{Z})^2 = (k-1)v + \sum(m_i-\bar{m})^2$. With $v = \sigma^2/n$, $m_i = \mu+\alpha_i$ and $\bar{m} = \mu$ (as $\sum\alpha_i = 0$),

$$E(\mathrm{SS}_{tr}) = n\Big\{(k-1)\frac{\sigma^2}{n} + \sum\alpha_i^2\Big\} = (k-1)\sigma^2 + n\sum\alpha_i^2.$$

Step 3. Distribution. $\mathrm{SSE}/\sigma^2 = \sum_i(n-1)S_i^2/\sigma^2$ is a sum of k independent $\chi^2(n-1)$ variables, hence $\chi^2(N-k)$. Each S_i^2 is independent of $\bar{Y}_{i\cdot}$ (normal sampling theory), and the groups are independent, so SSE is independent of $(\bar{Y}_{1\cdot}, \ldots, \bar{Y}_{k\cdot})$, hence of SS_{tr}. Under H_0 the $\bar{Y}_{i\cdot}$ are i.i.d. $N(\mu, \sigma^2/n)$, so $\mathrm{SS}_{tr}/\sigma^2 = \sum(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2/(\sigma^2/n) \sim \chi^2(k-1)$. Therefore $F = \dfrac{(\mathrm{SS}_{tr}/\sigma^2)/(k-1)}{(\mathrm{SSE}/\sigma^2)/(N-k)} \sim F(k-1, N-k)$.
■

Thus MSE always estimates σ^2, while MS_{tr} is inflated by treatment effects — which is why a large F indicates H_1.

Definition (Test of treatment effects (one-way, equal replication))

For $H_0 : \alpha_1 = \cdots = \alpha_k = 0$ use $F = \text{MS}_{tr}/\text{MSE}$. With observed value f:
(i) level-α critical region: $f \geq F_\alpha(k-1, N-k)$;
(ii) P-value: $P = P(F \geq f)$, $F \sim F(k-1, N-k)$.

Source	SS	d.f.	MS	F	P
Treatment	$\text{SS}_{tr} = n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	$k-1$	$\text{MS}_{tr} = \frac{\text{SS}_{tr}}{k-1}$	$f = \frac{\text{MS}_{tr}}{\text{MSE}}$	$P(F \geq f)$
Residual	$\text{SSE} = \sum_i\sum_j(y_{ij} - \bar{y}_{i\cdot})^2$	$N-k$	$\text{MSE} = \frac{\text{SSE}}{N-k}$		
Total	$\text{SST} = \sum_i\sum_j(y_{ij} - \bar{y}_{\cdot\cdot})^2$	$N-1$			

Example 9-1 (Concentration of a treatment solution)

Four concentrations of a fabric treatment (A_1: 3.0%, A_2: 3.5%, A_3: 4.0%, A_4: 4.5%) were each run 5 times in random order. Build the ANOVA table and test at level 5% whether concentration affects tensile strength.

	A_1	A_2	A_3	A_4
	47	51	50	22
	58	62	38	23
	51	31	47	28
	61	46	27	42
	46	49	23	25
mean	52.6	47.8	37.0	28.0

grand mean 41.35

Solution. Model $Y_{ij} = \mu + \alpha_i + e_{ij}$ with $k = 4$, $n = 5$; $H_0 : \alpha_1 = \cdots = \alpha_4 = 0$.

$$\text{SST} = (47 - 41.35)^2 + \cdots + (25 - 41.35)^2 = 3334.55,$$
$$\text{SS}_{tr} = 5\{(52.6 - 41.35)^2 + \cdots + (28.0 - 41.35)^2\} = 1826.55,$$
$$\text{SSE} = (47 - 52.6)^2 + \cdots + (25 - 28.0)^2 = 1508.00,$$

with d.f. 19, 3, 16. So $\text{MS}_{tr} = 608.85$, $\text{MSE} = 94.25$, $f = 6.46$, and $P = P(F(3,16) \geq 6.46) = 0.0045$ (table: between 0.001 and 0.01). Concentration has a significant effect on strength at level 5%.

Source	SS	d.f.	MS	F	P
Treatment	1826.55	3	608.85	6.46	0.0045
Residual	1508.00	16	94.25		
Total	3334.55	19			

9.2.2 Unequal replication

In practice the number of replicates often varies by treatment (e.g. more runs for a treatment of special interest). The analysis is almost the same. With n_i replicates of treatment i and $N = \sum n_i$,

$$\bar{y}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}, \qquad \mu = \frac{1}{N}\sum_{i=1}^{k} n_i\mu_i,$$

and $\alpha_i = \mu_i - \mu$ now satisfies $\sum n_i \alpha_i = 0$.

Definition (One-way ANOVA model (unequal replication))

$Y_{ij} = \mu + \alpha_i + e_{ij}$, $e_{ij} \sim N(0, \sigma^2)$ independent, $\sum_{i=1}^{k} n_i \alpha_i = 0$ $(i = 1, \dots, k,\ j = 1, \dots, n_i)$.

Proposition (Decomposition and expected mean squares, unequal replication)

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_{i=1}^{k} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with d.f. $N-1 = (k-1)+(N-k)$. Moreover $E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_{tr}) = \sigma^2 + \sum n_i \alpha_i^2/(k-1)$, and under $H_0 : \alpha_1 = \cdots = \alpha_k = 0$, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. The cross term is $2\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$, exactly as before; the middle square summed over j gives $n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Step 2. $E(\mathrm{SSE}) = \sum(n_i - 1)\sigma^2 = (N-k)\sigma^2$ as before.

Step 3. Write $\mathrm{SS}_{tr} = \sum n_i \bar{Y}_{i\cdot}^2 - N\bar{Y}_{\cdot\cdot}^2$. Since $E\bar{Y}_{i\cdot}^2 = \sigma^2/n_i + \mu_i^2$ and $\bar{Y}_{\cdot\cdot} \sim N(\mu, \sigma^2/N)$,

$$E(\mathrm{SS}_{tr}) = k\sigma^2 + \sum n_i \mu_i^2 - \sigma^2 - N\mu^2 = (k-1)\sigma^2 + \sum n_i(\mu_i - \mu)^2,$$

using $\sum n_i \mu_i = N\mu$. This gives $E(\mathrm{MS}_{tr})$.

Step 4. The independence argument of the equal-replication case is unchanged. Under H_0, $\sqrt{n_i}(\bar{Y}_{i\cdot} - \mu)/\sigma$ are i.i.d. $N(0,1)$, and $\mathrm{SS}_{tr}/\sigma^2$ is the squared length of the projection of this vector onto the $(k-1)$-dimensional subspace orthogonal to $(\sqrt{n_1}, \dots, \sqrt{n_k})$, hence $\chi^2(k-1)$. The F law follows. ■

Definition (Test of treatment effects (one-way, unequal replication))

$F = \mathrm{MS}_{tr}/\mathrm{MSE}$ with $\mathrm{MS}_{tr} = \mathrm{SS}_{tr}/(k-1)$, $\mathrm{MSE} = \mathrm{SSE}/(N-k)$, $N = \sum n_i$. Reject H_0 at level α if $f \geq F_\alpha(k-1, N-k)$; $P = P(F \geq f)$. The ANOVA table is as before with $\mathrm{SS}_{tr} = \sum_i n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Example 9-2 (Ratings on three review sites)

A film's ratings (out of 10) were sampled randomly: 4 from site 1, 5 from site 2, 6 from site 3. Test at level 5% whether the ratings differ by site.

	ratings	mean
Site 1	2, 3, 4, 5	3.5
Site 2	4, 5, 6, 4, 3	4.4
Site 3	6, 5, 7, 4, 6, 8	6.0

grand mean 4.8

Solution. $k = 3$, $n_1 = 4$, $n_2 = 5$, $n_3 = 6$; $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$.

$$\mathrm{SST} = (2-4.8)^2 + \cdots + (8-4.8)^2 = 36.4,$$
$$\mathrm{SS}_{tr} = 4(3.5-4.8)^2 + 5(4.4-4.8)^2 + 6(6.0-4.8)^2 = 16.2,$$
$$\mathrm{SSE} = (2-3.5)^2 + \cdots + (8-6.0)^2 = 20.2,$$

with d.f. 14, 2, 12. $\mathrm{MS}_{tr} = 8.1$, $\mathrm{MSE} = 1.68$, $f = 4.82$, $P = P(F(2,12) \geq 4.82) = 0.0292$. The ratings differ significantly by site.

Source	SS	d.f.	MS	F	P
Treatment	16.2	2	8.1	4.82	0.0292
Residual	20.2	12	1.68		
Total	36.4	14			

9.3 Two-way analysis of variance without replication

Motivation. A two-way layout examines two factors at once; with one observation per cell this requires no interaction.

Two-way ANOVA studies factors A $(A_1, \dots, A_p)$ and B $(B_1, \dots, B_q)$; the pq level combinations are the treatments, run in random order (running through B at fixed A violates randomization). All factors are fixed. With one observation y_{ij} per cell, define

$$\bar{y}_{i\cdot} = \frac{1}{q}\sum_{j=1}^{q} y_{ij}, \qquad \bar{y}_{\cdot j} = \frac{1}{p}\sum_{i=1}^{p} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{pq}\sum_{i=1}^{p}\sum_{j=1}^{q} y_{ij}.$$

Definition (Two-way ANOVA model (no replication))

$$Y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{p}\alpha_i = 0, \quad \sum_{j=1}^{q}\beta_j = 0,$$

$i = 1, \dots, p$, $j = 1, \dots, q$; α_i is the effect of level A_i and β_j that of B_j.

The hypotheses of interest are

$$H_0 : \alpha_1 = \cdots = \alpha_p = 0 \ \text{ vs. } \ H_1 : \text{some } \alpha_i \neq 0; \qquad H_0 : \beta_1 = \cdots = \beta_q = 0 \ \text{ vs. } \ H_1 : \text{some } \beta_j \neq 0.$$

The total deviation decomposes as

$$y_{ij} - \bar{y}_{\cdot\cdot} = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}),$$

into the deviation due to A_i, that due to B_j, and the residual.

Proposition (Decomposition, two-way without replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SSE}$, where

$$\mathrm{SST} = \textstyle\sum_i\sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 \ \ (pq-1), \quad \mathrm{SS}_A = q\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 \ \ (p-1),$$
$$\mathrm{SS}_B = p\sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 \ \ (q-1), \quad \mathrm{SSE} = \textstyle\sum_i\sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \ \ ((p-1)(q-1)),$$

with degrees of freedom in parentheses.

Proof. [Supplement] Write $a_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $b_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Then $\sum_i a_i = \sum_j b_j = 0$, and for each fixed i, $\sum_j r_{ij} = q\bar{y}_{i\cdot} - q\bar{y}_{i\cdot} - q\bar{y}_{\cdot\cdot} + q\bar{y}_{\cdot\cdot} = 0$; likewise $\sum_i r_{ij} = 0$ for each j.

Step 1. $\sum_i\sum_j a_i b_j = (\sum_i a_i)(\sum_j b_j) = 0$.

Step 2. $\sum_i\sum_j a_i r_{ij} = \sum_i a_i \sum_j r_{ij} = 0$ and $\sum_i\sum_j b_j r_{ij} = \sum_j b_j \sum_i r_{ij} = 0$.

Step 3. Squaring $y_{ij} - \bar{y}_{\cdot\cdot} = a_i + b_j + r_{ij}$ and summing, all cross terms vanish, leaving $q\sum a_i^2 + p\sum b_j^2 + \sum\sum r_{ij}^2$. The residuals satisfy $p+q-1$ independent linear constraints, so SSE has $pq - (p+q-1) = (p-1)(q-1)$ d.f.; and $(pq-1) = (p-1) + (q-1) + (p-1)(q-1)$. ∎

Mean squares and F-statistics:

$$\mathrm{MS}_A = \frac{\mathrm{SS}_A}{p-1}, \quad \mathrm{MS}_B = \frac{\mathrm{SS}_B}{q-1}, \quad \mathrm{MSE} = \frac{\mathrm{SSE}}{(p-1)(q-1)}, \qquad F_A = \frac{\mathrm{MS}_A}{\mathrm{MSE}}, \quad F_B = \frac{\mathrm{MS}_B}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way without replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \frac{q}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \frac{p}{q-1}\sum \beta_j^2$. Under the respective null hypotheses $F_A \sim F(p-1,(p-1)(q-1))$ and $F_B \sim F(q-1,(p-1)(q-1))$.

Proof. [Supplement]

Step 1. $\bar{Y}_{i\cdot} = \mu + \alpha_i + \bar{e}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and the $\bar{e}_{i\cdot}$ are independent $N(0,\sigma^2/q)$. As in the one-way proof, $E\sum_i(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2 = (p-1)\sigma^2/q + \sum \alpha_i^2$, so $E(\mathrm{SS}_A) = (p-1)\sigma^2 + q\sum \alpha_i^2$. Symmetrically for B.

Step 2. $\mathrm{SST} = \sum\sum(Y_{ij}-\bar{Y}_{\cdot\cdot})^2$ has expectation $(pq-1)\sigma^2 + \sum\sum(\alpha_i+\beta_j)^2 = (pq-1)\sigma^2 + q\sum\alpha_i^2 + p\sum\beta_j^2$ (the cross term vanishes by the constraints). Subtracting, $E(\mathrm{SSE}) = (pq-1-(p-1)-(q-1))\sigma^2 = (p-1)(q-1)\sigma^2$.

Step 3. The F laws follow from Cochran's theorem: $\mathrm{SS}_A, \mathrm{SS}_B, \mathrm{SSE}$ are quadratic forms in the errors with ranks $p-1, q-1, (p-1)(q-1)$ adding to $pq-1$, so under the null hypotheses they are independent $\sigma^2\chi^2$ variables with these d.f. ■

Definition (Tests of treatment effects (two-way, no replication))

(i) Factor A: $H_0 : \alpha_1 = \cdots = \alpha_p = 0$; $F_A = \mathrm{MS}_A/\mathrm{MSE}$; reject if $f_A \geq F_\alpha(p-1,(p-1)(q-1))$; $P_A = P(F_A \geq f_A)$.
(ii) Factor B: $H_0 : \beta_1 = \cdots = \beta_q = 0$; $F_B = \mathrm{MS}_B/\mathrm{MSE}$; reject if $f_B \geq F_\alpha(q-1,(p-1)(q-1))$; $P_B = P(F_B \geq f_B)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Residual	SSE	$(p-1)(q-1)$	MSE		
Total	SST	$pq-1$			

Example 9-3 (Diet and exercise)

Twenty similar men in their twenties were randomly assigned to one of 5 diets $(A_1,\ldots,A_5)$ and one of 4 exercises (B_1 running, B_2 weights, B_3 swimming, B_4 yoga). Weight loss (kg) after 4 weeks:

	A_1	A_2	A_3	A_4	A_5	mean
B_1	7.9	7.2	5.1	5.8	6.8	6.56
B_2	7.5	6.6	4.8	5.6	6.5	6.20
B_3	6.9	6.4	4.4	5.4	6.4	5.90
B_4	6.5	6.2	4.1	4.5	5.8	5.42
mean	7.20	6.60	4.60	5.33	6.38	6.02

Test at level 5% whether diet and exercise affect weight loss.
Solution. $H_0 : \alpha_1 = \cdots = \alpha_5 = 0$ and $H_0 : \beta_1 = \cdots = \beta_4 = 0$.

$$\mathrm{SS}_A = 4\{(7.20-6.02)^2 + \cdots + (6.38-6.02)^2\} = 17.417,$$

$$\mathrm{SS}_B = 5\{(6.56-6.02)^2+\cdots+(5.42-6.02)^2\} = 3.492,$$
$$\mathrm{SST} = (7.9-6.02)^2+\cdots+(5.8-6.02)^2 = 21.232,$$
$$\mathrm{SSE} = \mathrm{SST}-\mathrm{SS}_A-\mathrm{SS}_B = 0.323,$$

with d.f. $4, 3, 19, 12$. Then $\mathrm{MS}_A = 4.354$, $\mathrm{MS}_B = 1.164$, $\mathrm{MSE} = 0.027$, $f_A = 161.77$, $f_B = 43.24$, both with $P < 0.000001$. Both the diet and the exercise type have significant effects at level 5%.

Source	SS	d.f.	MS	F	P
Factor A	17.417	4	4.354	161.77	< 0.000001
Factor B	3.492	3	1.164	43.24	< 0.000001
Residual	0.323	12	0.027		
Total	21.232	19			

9.4 Two-way analysis of variance with replication

Motivation. Replication separates an *interaction*, an effect of A varying with B, from pure error.

Each of the pq combinations is replicated r times, all pqr runs in random order (running the r replicates of each combination consecutively is not completely randomized).

Replication lets us estimate the **interaction**: the effect of A depends on the level of B. Without it, $E(Y_{ij}) = \mu + \alpha_i + \beta_j$, so the difference between two levels of A is the same at every level of B, e.g.

$$(\mu+\alpha_1+\beta_j)-(\mu+\alpha_3+\beta_j) = \alpha_1-\alpha_3 \quad \text{for all } j.$$

Plotting mean response against B, one line per level of A, gives parallel segments without interaction (Figure 1, left) and crossing lines with interaction (right), when the unreplicated analysis fails.

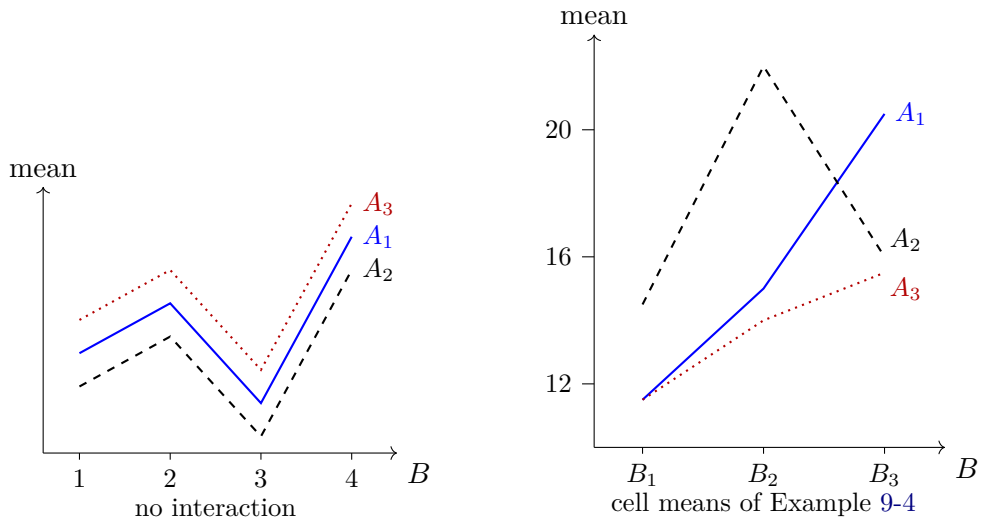


Figure 1: Interaction plots. Left: parallel profiles (no interaction). Right: crossing profiles, here the cell means $\bar{y}_{ij\cdot}$ of Example 9-4.

The data are y_{ijk} $(i = 1,\ldots,p;\ j = 1,\ldots,q;\ k = 1,\ldots,r)$ with means

$$\bar{y}_{ij\cdot} = \frac{1}{r}\sum_k y_{ijk}, \quad \bar{y}_{i\cdot\cdot} = \frac{1}{qr}\sum_j\sum_k y_{ijk}, \quad \bar{y}_{\cdot j\cdot} = \frac{1}{pr}\sum_i\sum_k y_{ijk}, \quad \bar{y}_{\cdots} = \frac{1}{pqr}\sum_i\sum_j\sum_k y_{ijk}.$$

Let γ_{ij} be the interaction effect of A_i and B_j.

Definition (Two-way ANOVA model (with replication))

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad e_{ijk} \sim N(0, \sigma^2) \text{ independent},$$
$$\sum_{i=1}^{p} \alpha_i = 0, \quad \sum_{j=1}^{q} \beta_j = 0, \quad \sum_{i=1}^{p} \gamma_{ij} = \sum_{j=1}^{q} \gamma_{ij} = 0,$$

$i = 1, \dots, p$, $j = 1, \dots, q$, $k = 1, \dots, r$.

Tests of the main effects proceed as in §9.3; the new question is the interaction:

$$H_0 : \gamma_{ij} = 0 \ \ (\text{all } i, j) \qquad \text{vs.} \qquad H_1 : \text{at least one } \gamma_{ij} \neq 0.$$

The total deviation decomposes as

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}),$$

into deviations due to A_i, to B_j, to the interaction, and the residual.

Proposition (Decomposition, two-way with replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B} + \mathrm{SSE}$, where

$$\begin{aligned}
\mathrm{SST} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{\cdots})^2 && (pqr - 1),\\
\mathrm{SS}_A &= qr \textstyle\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 && (p - 1),\\
\mathrm{SS}_B &= pr \textstyle\sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 && (q - 1),\\
\mathrm{SS}_{A\times B} &= r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 && ((p-1)(q-1)),\\
\mathrm{SSE} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2 && (pq(r-1)).
\end{aligned}$$

Proof. [Supplement]

Step 1. Split $y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot})$. This is a one-way layout with the pq cells as treatments, each replicated r times, so by the one-way decomposition

$$\mathrm{SST} = r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 + \mathrm{SSE}.$$

Step 2. The cell means $\bar{y}_{ij\cdot}$ form a $p \times q$ table whose row means are $\bar{y}_{i\cdot\cdot}$, column means $\bar{y}_{\cdot j\cdot}$ and grand mean $\bar{y}_{\cdots}$. Applying the two-way decomposition of §9.3 to this table and multiplying by r,

$$r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B}.$$

Step 3. Combining gives the identity. D.f.: $pq(r-1)$ for SSE ($r-1$ per cell), and $pq - 1 = (p-1) + (q-1) + (p-1)(q-1)$ for the cell-mean part; the total is $pqr - 1$. ∎

Mean squares and the interaction test statistic:

$$\mathrm{MS}_{A\times B} = \frac{\mathrm{SS}_{A\times B}}{(p-1)(q-1)}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{pq(r-1)}, \qquad F = \frac{\mathrm{MS}_{A\times B}}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way with replication)

$E(\text{MSE}) = \sigma^2$, $E(\text{MS}_A) = \sigma^2 + \dfrac{qr}{p-1}\sum \alpha_i^2$, $E(\text{MS}_B) = \sigma^2 + \dfrac{pr}{q-1}\sum \beta_j^2$, and $E(\text{MS}_{A\times B}) = \sigma^2 + \dfrac{r}{(p-1)(q-1)}\sum\sum \gamma_{ij}^2$. Under $H_0 : \gamma_{ij} = 0$, $F = \text{MS}_{A\times B}/\text{MSE} \sim F((p-1)(q-1), pq(r-1))$.

Proof. [Supplement]

Step 1. Within each cell, $Y_{ij1}, \dots, Y_{ijr}$ is a sample from $N(\mu_{ij}, \sigma^2)$ with $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, so $E(\text{SSE}) = pq(r-1)\sigma^2$ and $\text{SSE}/\sigma^2 \sim \chi^2(pq(r-1))$, independent of all cell means (normal sampling theory, cells independent).

Step 2. The cell means $\bar{Y}_{ij\cdot} = \mu_{ij} + \bar{e}_{ij\cdot}$ are independent with variance σ^2/r, i.e. a two-way table without replication with error variance σ^2/r and "means" $\mu+\alpha_i+\beta_j+\gamma_{ij}$. By the computation in §9.3 (now with the extra term γ_{ij}, whose row and column sums vanish), $E(\text{SS}_A) = (p-1)\sigma^2 + qr\sum \alpha_i^2$, $E(\text{SS}_B) = (q-1)\sigma^2 + pr\sum \beta_j^2$ and

$$E(\text{SS}_{A\times B}) = (p-1)(q-1)\sigma^2 + r\textstyle\sum_i\sum_j \gamma_{ij}^2.$$

Step 3. $\text{SS}_{A\times B}$ is a function of the cell means, hence independent of SSE; under H_0 it is $\sigma^2\chi^2((p-1)(q-1))$ (Cochran's theorem applied to the cell-mean table, as in §9.3). The F law follows; the main-effect F-tests are justified the same way. ■

Definition (Test of interaction (two-way with replication))

For $H_0 : \gamma_{ij} = 0$ $(i = 1, \dots, p,\ j = 1, \dots, q)$ use $F = \text{MS}_{A\times B}/\text{MSE}$; reject at level α if $f \geq F_\alpha((p-1)(q-1), pq(r-1))$; $P = P(F \geq f)$. The main effects are tested by $F_A = \text{MS}_A/\text{MSE}$ and $F_B = \text{MS}_B/\text{MSE}$ with denominator d.f. $pq(r-1)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \text{MS}_A/\text{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \text{MS}_B/\text{MSE}$	$P(F_B \geq f_B)$
Interaction	$\text{SS}_{A\times B}$	$(p-1)(q-1)$	$\text{MS}_{A\times B}$	$f_{A\times B} = \text{MS}_{A\times B}/\text{MSE}$	$P(F_{A\times B} \geq f_{A\times B})$
Residual	SSE	$pq(r-1)$	MSE		
Total	SST	$pqr-1$			

Example 9-4 (Machines and operators)

Three machines (A_1, A_2, A_3) and three operators (B_1, B_2, B_3) were compared by measuring productivity twice for each combination. Test the main effects and the interaction at level 5%.

	B_1	B_2	B_3	mean
A_1	9, 14	14, 16	19, 22	15.67
A_2	13, 16	18, 26	14, 18	17.5
A_3	11, 12	11, 17	15, 16	13.67
mean	12.5	17	17.33	15.61

Solution. Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$; $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$; $H_0 : \gamma_{ij} = 0$ $(i, j = 1, 2, 3)$.

$$\text{SS}_A = 3 \times 2\{(15.67 - 15.61)^2 + \cdots + (13.67 - 15.61)^2\} = 44.11,$$

$$\begin{aligned}
\mathrm{SS}_B &= 3 \times 2\{(12.5 - 15.61)^2 + \cdots + (17.33 - 15.61)^2\} = 87.44,\\
\mathrm{SS}_{A\times B} &= 2\{(11.5 - 15.67 - 12.5 + 15.61)^2 + \cdots\} = 74.22,\\
\mathrm{SST} &= (9 - 15.61)^2 + \cdots + (16 - 15.61)^2 = 288.28,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B - \mathrm{SS}_{A\times B} = 82.51,
\end{aligned}$$

with d.f. $2, 2, 4, 17, 9$. Hence $\mathrm{MS}_A = 22.06$, $\mathrm{MS}_B = 43.72$, $\mathrm{MS}_{A\times B} = 18.56$, $\mathrm{MSE} = 9.17$ and $f_A = 2.41$, $f_B = 4.77$, $f_{A\times B} = 2.02$, with P-values 0.146, 0.039, 0.174. Operators differ significantly in productivity at level 5%, while the machine type and the interaction are not significant.

Source	SS	d.f.	MS	F	P
Factor A	44.11	2	22.06	2.41	0.146
Factor B	87.44	2	43.72	4.77	0.039
Interaction	74.22	4	18.56	2.02	0.174
Residual	82.51	9	9.17		
Total	288.28	17			

Model checking and beyond

Check the error assumptions first by plotting studentized residuals against estimated means; for Example 9-4 no problem appears. Estimating or comparing level means after a significant result is beyond these notes.